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Project Title:	Effects of Microgravity on Preimplantation Mouse Development		
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Task Description:	A truism in mammalian development is that fertilization is not difficult, but implantation is. About 70% of all human embryos die before birth, and the time of greatest embryonic loss is during implantation. (Cross et al., 1994). In this period, the embryonic, and extraembryonic yolk sac and placental cell lineages are determined and become essential for survival (Rappolee, 1999). In humans, failure to form a placenta properly leads to the clinically important conditions preeclampsia and intrauterine growth retardation (IUGR). Experiments have shown that perturbation of maternal or embryonic development only during the late preimplantation stages can lead to detrimental ramifications (Rutedge et al., 1994) during later development and at parturition (Kwong et al., 2000). Hypothesis 1: Early cleavage division events are sensitive to gravity. Hypothesis 1a, the gravity-sensitive period is only during the 2-8-cell-stage when the block in embryonic development was observed (Kojima et al., 2000), and post-8-cell development is normal (if embryos escape the block). Hypothesis 1b, the gravity-sensitive period also occurs during post-8-cell development after compaction. Specific aim 1. (1) Which stage of preimplantation embryonic development is gravity-sensitive? To test this, we will make daily measurements of several criteria of embryo development: (1) anabolic deviation (RNA, protein content and synthetic rates), (2) net growth rate (cell number), (3) components contributing to net growth rate (fraction in mitosis S phase, fraction in apoptosis), (4) morphogenetic events (compaction rate, cavitation rate), (5) and allocation to embryonic and placental lineages (fraction and total number inside and outside cells). Analysis of preimplantation development will be done to distinguish whether pre-compaction (2-8-cell-stage) development that is the primary period of perturbation (Kojima et al., 2000). What events are affected by microgravity? Zygotic genome activation (ZGA), compaction and cavitation will be tested for delay of initiation and quantitatively for relative strength. This will distinguish whether events during the 2-8-cell-stage block (Kojima et al., 2000) have an earlier onset. Post-compaction events such as cavitation and trophoblast lineage allocation will be tested for indication that development of the embryo is perturbed after the 2-8-cell-stage block-embryo. Hypothesis 2: Preimplantation embryos are sensitive to gravity and embryos cultured under conditions of microgravity during the (1) entire preimplantation development, (2) from 1-8-cell- stage, or (3) from post-compaction 8-cell-stage through blastocyst-stage show gravity-sensitive phenotypes. These phenotypes are observed in two post-implantation models; (1) when embryos from these three groups produce smaller litter sizes/weights after reimplantation and (2) when they produce deviant early post-implantation events measured in blastocyst or ICM outgrowth culture. Specific aim 2 (2) Are embryos that progress to the most advanced preimplantation stages (morula and blastocyst) in microgravity culture normal when reimplanted or cultured? (2a) Do embryos cultured in microgravity from (1) 1-cell-stage through blastocyst-stage, (2) from 8-cell-stage through blastocyst-stage, or (3) from 1-cell-stage through 8-cell-stage reimplant and produce normal quantity of newborns? Analyze number and weight of newborn mice produced from embryos culture in microgravity for parts or all of preimplantation development. If number or weight is decreased, do follow-up studies on size of embryos during mid- and late-uterine growth periods. (2b) Do blastocyst cultured in microgravity (1) from 1-cell to blastocyst-stage, or from 8-cell-stage through blastocyst-stage yield the normal placental/trophoblast and yolk sac/endoderm growth and differentiation events in blastocyst outgrowth culture at normal gravity? Analysis of blastocyst outgrowth culture using markers for trophoblast and endoderm proliferation, differentiation and apoptosis will be done to test for effects of microgravity on preimplantation embryos. Determine whether simulated microgravity or shear stress influence early mammalian embryonic development between fertilization and the implantation of the embryo into the uterus. Will determine whether mammals can successfully undergo early embryonic development in low gravity (simulated). Additional benefits will include the definition of stress by measuring, defining stress reactions, and blocking stress reactions, under conditions of microgravity, shear stress and stress during normal culture used in IVF clinics.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will inform us whether gravity has a role during mammalian embryogenesis as the embryo prepares to set up the head-to-tail, back-to-belly, and left-to-right axes. Rapid lethality of embryos due to MGS and shear stress was much more rapid than anticipated and positional morphogenetic information was not tested. In addition, we will test whether embryonic movement set up during microgravity simulation is itself stressful and deleterious to embryogenesis before implantation into the uterus. The analysis of stress caused by microgravity simulation, shear stress, and stress during embryo handling and culture has led to many benefits. A patent is in process to use stress enzyme inhibitors to improve embryo health, and the health of the offspring of IVF (approaching two million offspring). Further diagnostic and prognostic reagents, and information on how to handle embryos without stress, have come from these studies.
	Laboratory progress, project completion, and future studies. 2004-2006. Previous work. Year three (2004). Dr Wang concluded her study by finding that E1.5 (2-cell stage) was as sensitive to MGS caused lethality as E3.5, but that E2.5 (8-cell stage) was the most sensitive to MGS-induced lethality in MGS>RC>>>>SC. The 8-cell stage was the most sensitive time period of development, but RC was also very lethal at this stage. This is the period when the first placental epithelium forms and is known to be an important developmental stage. We anticipated that there might be slow deviations in morphological determination, but instead a fast cell suicide and embryo death was observed. Even when embryos were placed in recovery culture after MGS for 24hr, they still died. Therefore, instead of testing for slow changes in markers of differentiation, we examined fast induction of stress response enzymes such as stress activated protein kinase and p38 mitogen-activated protein kinase (MAPK). Since these were not previously studied in early mouse embryos, we needed to develop an understanding of these enzymes. A graduate student in the lab (Dr Zhong) showed that SAPK and p38MAPK were expressed throughout early mouse embryonic development, and in early mouse and human placental cells (Zhong et al., 2004). Dr Wang studied the relationship of phosphorylated SAPK and p38MAPK in preimplantation embryo development. Dr Wang also found that SAPK phosphorylation was associated with embryo death caused by MGS. Dr Wang left the lab for a second post-doctoral fellowship and Dr Xie took over the project. Dr Xie established an RCCS1 rotation similar to Dr Wang's, but used larger aqueous media droplets that did not orbit in the oil as the vessel rotated at 7.5 RPM. This created more shear stress (SS) and minimal MGS. E3.5 embryos became

apoptotic and died within 12hr or SS, suggesting a more rapid lethality than orbital MGS that approximated 0.1% gravity.

SAPK/JNK phosphorylation preceded the TUNEL and rapid lethality of MGS and slower lethality of the RC group. A SAPK/JNK inhibitor significantly reduced the fraction of TUNEL cells, suggesting that SAPK/JNK was causal in SS. Fos was quickly induced (0.5hr MGS), suggesting shear stress developed rapidly and was sensed by the outer placental epithelial cells. In 1970, Bronson and McLaren reported that removing the Zona pellucida (an acellular 4% protein coat around the oocyte preimplantation embryo) at the 8-cell stage was lethal to embryos that were reimplanted into the oviduct where they would move, but not in static embryos in culture (citation available upon request). We tested whether the Zona pellucida might protect the embryo from shear stress. Embryos with the Zona removed were significantly more susceptible to lethality induced by shear stress, but embryos without Zona were not significantly more TUNEL in static culture than embryos with Zona. Summary; MGS or shear stress leads to SAPK/JNK phosphorylation/activation that precedes lethality. SAPK/JNK inhibitors (SP600125 or D-JNKI1) can largely block shear stress-induced TUNEL/apoptosis, suggesting that SAPK activity is causal for apoptosis.

Year four (2005). Shear stress was quantified when the Synthecon RCCS1 HARV was run (7.5RPM) with a stationary media microdroplet at 1.2dynes/cm2, about 1/10th the amount of shear stress sensed by arteriolar endothelial cells. DV digital movies of the stationary droplet and the embryos inside the droplet were recorded and sent to our collaborator, Dr Shen (a NASA-funded laminar flow scientist), who solved a free body analysis of the motion and amount of shear stress on the embryos. This amount of shear stress is about 100x the average amount of shear stress sensed by the embryo during transit from the ovary through the oviduct, into the uterus. (Although Dr Croxatto reports that episodic maximal velocities in the oviduct may approach the velocity and shear observed in the RCCS1, personal communication). The acellular protein coat, the Zona pellucida protects the embryo from the 1-cell zygote immediately after fertilization (in the upper oviduct) until the blastocyst stage 4.5 days later (E4.5), when the embryo hatches from the Zona pellucida and implants into the uterus. Since the embryos attach and become stationary after attachment, while fluids in the uterine lumen continue to flow over the embryo surface, we measured SAPK/JNK phospho at E4.5 by Western blot and found it high. The removal of the Zona pellucida leads to a more rapid induction of embryo lethality, suggesting a function of the Zona pellucida is to protect the embryo during transit through the oviduct and uterus. C-Jun phospho is also upregulated during shear stress, suggesting that both components of the AP-1 transcription-regulating heterodimer is upregulated by shear. We also found that mouth pipetting embryos increases both SAPK/JNK phospho and c-Fos in a dose-dependent day.

Another variable arising during shear stress or microgravity simulation due to movement, is the loss of hypoxia. Hypoxia is a known mitogen to the placental stem cells in the outer epithelium of the preimplantation embryo. However, when the RCCS1/HARV protocols were done in a hypoxic incubator, the shear stress and microgravity-simulated groups were not affected (although there was a significant improvement in embryo development and lessened lethality in rotational controls).

That shear stress might lead to activation of the essential “implantation phenotype” has recently been tested. The implanting embryos must differentiate (placental) trophoblast stem (TS) cells to make trophoblast giant (TG) cells that signal directly to the mother during early gestation that the embryo is present. This signaling maintains the corpus luteum of the ovary and maintains progesterone section by the corpus luteum essential for embryonic survival. The signal is hCG in human (ligand of the early pregnancy test), and placental lactogen (PL)-1 in the mouse. During differentiation, Inhibitor of Differentiation (Id)-2 (basic Helix loop helix dominant negative) must be suppressed and Heart and mesoderm differentiation inducer (HAND1, basic helix loop helix transcription factor that regulates the promoter of PL-1) must be stabilized. We have found that stress on the E3.5 embryo leads to fast Id2 export from the nucleus of TS cells, leaving only HAND1 in the nucleus. In TS cells, (hyperosmolar) stress causes Id2 export and then loss of total protein in a SAPK/JNK-independent manner. Stress also causes stabilization and increase in HAND1 in a SAPK/JNK-dependent manner. PL-1 is stress-induced in manner partially dependent on SAPK/JNK.

We will test for other stress enzymes that contribute to stress sensing in the outer placental cells of the implanting embryo and we are also testing for whether the shear stress regulates the induction of PL-1 in the implanting embryo after hatching and loss of the Zona Pellucida.

In this year, we published Dr Wang’s work that showed that levels of phosphorylated SAPK are inversely correlated with embryo development in static culture (Wang et al., 2005) and showed that elements of the essential lipophilic MAPK pathway (Xie et al., 2005b) and the mitogenic pathway it controls (Xie et al., 2005a) are expressed in early mouse embryos and can be controlled by stress and SAPK (data not shown).

Year five (2006), completion.

Work on the shear stress generation of cell death (and associated with embryo death), via SAPK and blocked by the Zona pellucida, was published (Xie et al., 2006c). An additional analysis of the role of SAPK during embryonic development of embryos in the most and least optimized media (least and most powerful in phosphorylating and activating SAPK, respectively) was also published (Xie et al., 2006a). Together these reports show that there are quantities and qualities of stress to embryos by poor media or shear stress that 1) are so high they lead to embryo death, 2) require SAPK for embryo homeostatic response to stress (but at some point stress levels may also klll embryos despite SAPK homeostasis), and 3) cause an over-reaction to stress that is detrimental to embryonic developmental. These findings have led to the filing of a patent for the use of stress enzyme inhibitors to improve embryo development in the optimized media currently use in in vitro fertilization (ICF) clinics. In addition, the development of shear stress induced by embryo handling during IVF procedures (dose-dependent pipetting) was also established by Dr Xie and has been submitted (Xie et al., 2006b). Both phosphorylated SAPK and Fos protein are induced by handling embryos with increased sensitivity (lower pipetting threshold) at E4.5 when the zona pellucida is lost. The paper on lethality generated by MGS has also been resubmitted (Wang et al., 2006). This study was limited in that it was not possible to empirically measure both SS and MGS in the MGS group. Circumstantial evidence suggests that the rapid induction of lethality compared with SS-only groups, and the calculated low SS in the MGS group, is due to MGS, not the residual small amounts of SS.

Future studies. Several studies are clearly needed.

Task Progress:

1. A dose response of shear stress is needed to understand the role of shear stress during normal development in the oviduct and uterus, after fertilization and before implantation of the embryo into the uterus. We are working on a model, with Dr Shen and the Synthecon scientists, using different viscosities and densities of oil, to create different SS levels in

the RCCS-1. This will enable us to predict the level (and duration) of stress that might lead to pathological consequences in normal development, IVF protocols, and in low gravity environments in space. A SS dose-response will also allow us to assign the relative contribution of MGS and SS to the lethality generated in MGS when SS levels are hypothetically very low.

2. It is important to test whether embryos can be fertilized, develop and implant in low gravity in Space. We are exploring the possibility of testing for the roles of low microgravity on the International Space Station with Dr van Loon of the Dutch microgravity simulation center (DESC/Dutch Experiment Support Center) and the European Space Agency.

3. In anticipation of a space-based microgravity test, further microgravity simulation effects on embryos can be tested. A method for determining the amount of MGS and of SS in these experiments must be perfected. Two possible ways to measure embryo movement in the MGS drop rotating with the RCCS-1 are possible. One is the use of a rapid stroboscopic illumination of a series of micrographs of the embryos in the rotating drop as it moves past a static stereomicroscope objective. The second is a stereomicroscope flexible fiberoptic extension on a rotating second RCCS1 that closely follows the microdroplet's rotation on the first RCCS1. Neither of these technologies have been shown to work although initial testing of each was tried during this grant.

4. It is important to determine if stress generated by shear or poor media, accelerate a premature differentiation of embryonic stem cells or placental stem cells that is detrimental to the long term health of the embryo, fetus, or offspring. We are currently testing whether stressed embryos undergo premature differentiation as we have found for stressed placental stem cells derived from the embryo. This work was supported by the following grants NASA (NAG2-1503), NIH (R01 HD40972A). Additional help was kindly given by Dr Josh Zimmerberg of NICHD, who provided 2 additional RCCS1 and many 10mL HARVS. These additional RCCS1s allowed us to control experiments while developing more data points/experiment for shear stress time courses and SAPK/JNK inhibitor studies.

Publications

*Three papers have been published in 2006 (Xie et al., 2006a; Xie et al., 2006c; Zhong et al., 2006), one directly comes from NASA-funded research (Xie et al., 2006c) and one was partially supported by NASA (Xie et al., 2006a). One paper has been resubmitted that comes directly out of the shear stress research performed in the Synthecon RCCS-1. A patent is also in preparation for the use of inhibition of SAPK and other stress enzymes in IVF media.

*Six papers were published by our lab in the 2004-5 progress report period (Liu et al., 2004; Wang et al., 2005; Wang et al., 2004; Xie et al., 2005a; Xie et al., 2005b; Zhong et al., 2004).

Liu, J., Puscheck, E. E., Wang, F., Zhong, W., Trostinskaia, A. B., Barisic, D., Maniere, G., Wygle, D., Rings, E. H. H., and Rappolee, D. A. (2004). Serine-threonine kinases and transcription factors active in signal transduction are detected at high levels of phosphorylation during mitosis in preimplantation embryos and trophoblast stem cells (TSC). Reproduction 128, 643-654.

Wang, Y., Puscheck, E. E., Wygle, D. L., Lewis, J. J., Trostinskaia, A. B., Wang, F., and Rappolee, D. A. (2005). Increases in phosphorylation of SAPK/JNK and p38MAPK correlate negatively with mouse embryo development after culture in different media. Fertility and Sterility. 83, 1144-1154.

Wang, Y., Wang, F., Sun, T., Trostinskaia, A., Wygle, D., Puscheck, E., and Rappolee, D. A. (2004). Entire mitogen activated protein kinase (MAPK) pathway is present in preimplantation mouse embryos. Dev Dyn 231, 72-87.

Wang, Y. C., Xie, Y., Wygle, D. L., Shen, H., Puscheck, E. E., and Rappolee, D. A. (2006). A major effect of simulated microgravity on preimplantation mouse development is lethality associated with elevated phosphorylated SAPK/JNK expression. Molecular Reproduction and Development Submitted.

Xie, Y., Puscheck, E. E., and Rappolee, D. A. (2006a). Effects of SAPK/JNK inhibitors on preimplantation mouse embryo development are influenced greatly by the amount of stress induced by the media. Molecular Human Reproduction. In press.

Xie, Y., Wang, F., Puscheck, E. E., and Rappolee, D. A. (2006b). Pipetting causes dose-dependent shear stress and elevation of SAPK/JNK phosphorylation in preimplantation mouse embryos. Molecular Reproduction and Development Submitted.

Xie, Y., Wang, F., Zhong, W., Puscheck, E. E., Shen, H., and Rappolee, D. A. (2006c). Shear Stress Induces Preimplantation Embryo Death That is Delayed By the Zona Pellucida and Associated with Stress-Activated Protein Kinase-mediated Apoptosis. Biology of Reproduction In press.

Xie, Y., Wang, T., Sun, T., Wang, Y., Wang, F., Puscheck, E. E., and Rappolee, D. A. (2005a). Acquisition of essential somatic cell cycle regulatory protein expression and implied activity occurs at the second to third cell division in mouse preimplantation embryos. FEBS Lett 579, 398-408.

Xie, Y., Wang, Y., Sun, T., Wang, F., Trostinskaia, A., Puscheck, E. E., and Rappolee, D. A. (2005b). Six post-implantation lethal knockouts of genes for lipophilic MAPK pathway proteins are expressed in preimplantation mouse embryos and trophoblast stem cells. Mol Reprod Dev 71, 1-11.

Zhong, W., Sun, T., Wang, Q., Wang, Y., Xie, Y., Johnson, A., Leach, R., Puscheck, E. E., and Rappolee, D. A. (2004). SAPK/JNK1, 2, but not SAPK/JNK3 mRNA transcripts, are expressed in early gestation human placenta and mouse eggs, preimplantation embryos, and trophoblast stem cells. Fertility and Sterility. 82, 1140-1148.

Zhong, W., Wang, Q. T., Sun, T., Wang, F., Liu, J., Leach, R., Johnson, A., Puscheck, E. E., and Rappolee, D. A. (2006). FGF ligand family mRNA expression profile for mouse preimplantation embryos, early gestation human placenta, and mouse trophoblast stem cells. Molecular Reproduction and Development 73, 540-550.

Meetings

Oral presentations

-- Great Lakes Mammalian Development conference, invited speaker and poster presentations, Toronto, CA, 2006.

-- "Downunder embryo". A Symposium on the Cell Biology of the Early Embryo, invited keynote speaker, Queensland

	Australia, 2006. Poster presentations -- Xie, Y, Wang, F, Zhong, W, Puscheck, EE, Shen, H, and DA Rappolee. (2005). Shear stress causes rapid lethality in preimplantation mouse embryos by induction of the SAPK-mediate apoptosis. A novel function of the Zona Pellucida is to prevent shear stress-associated lethality. ASRM, Montreal Quebec. -- Xie, Y, Wang, F, Zhong, W, Puscheck, EE, Shen, H, and DA Rappolee (2006). Shear stress is lethal To preimplantation mouse embryos by induction of SAPK-mediated apoptosis (TUNEL). A second function of the Zona Pelucida is to prevent shear stress-associated lethality. Great lakes mammalian development conference, Toronto, Ontario, Canada. -- Xie, Y, Wang, F, Puscheck, EE, and DA Rappolee (2006). Pipetting shear stress increases FOS protein and phosphorylated SAPK/JNK in a dose-dependent manner. Society for the study of reproduction (SSR), Omaha NE.
Bibliography Type:	Description: (Last Updated: 09/23/2009)
Articles in Peer-reviewed Journals	Xie Y, Wang F, Zhong W, Puscheck E, Shen H, Rappolee DA. "Shear stress induces preimplantation embryo death that is delayed by the zona pellucida and associated with stress-activated protein kinase-mediated apoptosis." Biol Reprod. 2006 Jul;75(1):45-55. PMID: 16571875 , Jul-2006
Articles in Peer-reviewed Journals	Xie Y, Puscheck EE, Rappolee DA. "Effects of SAPK/JNK inhibitors on preimplantation mouse embryo development are influenced greatly by the amount of stress induced by the media." Mol Hum Reprod. 2006 Apr;12(4):217-24. PMID: 16574700 , Apr-2006
Articles in Peer-reviewed Journals	Zhong W, Wang QT, Sun T, Wang F, Liu J, Leach R, Johnson A, Puscheck EE, Rappolee DA. "FGF ligand family mRNA expression profile for mouse preimplantation embryos, early gestation human placenta, and mouse trophoblast stem cells." Mol Reprod Dev. 2006 May;73(5):540-50. PMID: 16470835 , May-2006
Articles in Peer-reviewed Journals	Wang Y, Puscheck EE, Lewis JJ, Trostinskaia AB, Wang F, Rappolee DA. "Increases in phosphorylation of SAPK/JNK and p38MAPK correlate negatively with mouse embryo development after culture in different media." Fertil Steril. 2005 Apr;83 Suppl 1:1144-54. PMID: 15831287 , Apr-2005
Articles in Peer-reviewed Journals	Zhong W, Sun T, Wang QT, Wang Y, Xie Y, Johnson A, Leach R, Puscheck EE, Rappolee DA. "SAPKgamma/JNK1 and SAPKalpha/JNK2 mRNA transcripts are expressed in early gestation human placenta and mouse eggs, preimplantation embryos, and trophoblast stem cells." Fertil Steril. 2004 Oct;82 Suppl 3:1140-8. PMID: 15474087 , Oct-2004
Articles in Peer-reviewed Journals	Liu J, Puscheck EE, Wang F, Trostinskaia A, Barisic D, Maniere G, Wygle D, Zhong W, Rings EH, Rappolee DA. "Serine-threonine kinases and transcription factors active in signal transduction are detected at high levels of phosphorylation during mitosis in preimplantation embryos and trophoblast stem cells." Reproduction. 2004 Nov;128(5):643-54. PMID: 15509711 , Nov-2004
Articles in Peer-reviewed Journals	Rappolee DA. "Impact of transient stress and stress enzymes on development." Dev Biol. 2007 Apr 1;304(1):1-8. Review. PMID: 17258702 , Apr-2007
Articles in Peer-reviewed Journals	Xie Y, Zhong W, Wang Y, Trostinskaia A, Wang F, Puscheck EE, Rappolee DA. "Using hyperosmolar stress to measure biologic and stress-activated protein kinase responses in preimplantation embryos." Mol Hum Reprod. 2007 Jul;13(7):473-81. PMID: 17483086 , Jul-2007
Articles in Peer-reviewed Journals	Xie Y, Wang F, Puscheck EE, Rappolee DA. "Pipetting causes shear stress and elevation of phosphorylated stress-activated protein kinase/jun kinase in preimplantation embryos." Mol Reprod Dev. 2007 Oct;74(10):1287-94. PMID: 17492777 , Oct-2007
Articles in Peer-reviewed Journals	Xie Y, Liu J, Proteasa S, Proteasa G, Zhong W, Wang Y, Wang F, Puscheck EE, Rappolee DA. "Transient stress and stress enzyme responses have practical impacts on parameters of embryo development, from IVF to directed differentiation of stem cells." Mol Reprod Dev. 2008 Apr;75(4):689-97. PMID: 17654540 , Apr-2008
Articles in Peer-reviewed Journals	Zhong W, Xie Y, Wang Y, Lewis J, Trostinskaia A, Wang F, Puscheck EE, Rappolee DA. "Use of hyperosmolar stress to measure stress-activated protein kinase activation and function in human HTR cells and mouse trophoblast stem cells." Reprod Sci. 2007 Sep;14(6):534-47. PMID: 17959882 , Sep-2007
Articles in Peer-reviewed Journals	Wang Y, Xie Y, Wygle D, Shen HH, Puscheck EE, Rappolee DA. "A major effect of simulated microgravity on several stages of preimplantation mouse development is lethality associated with elevated phosphorylated SAPK/JNK." Reproductive Sciences. 2009 Oct;16(10):947-59. PubMed PMID: 19546324 , Oct-2009